

Rank of a Matrix

From Systems of Statements to Row Echelon Form — and Why It Matters



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$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{bmatrix}$$

What We'll Cover

1

Motivation

Systems of statements — complete, redundant, or contradictory?

2

Definition & Theory

Rank, row echelon form, and elementary row operations

3

Worked Example

Finding the rank of a 3×3 matrix step by step

4

Real-World Application

How rank reduction compresses digital images

Warm-Up: Systems of Statements

Before we touch a single number, let's build intuition with sentences instead of equations.

System 1

- The dog is **black**
- The cat is **orange**

Complete

Non-singular

System 2

- The dog is **black**
- The dog is **black**

Redundant

Singular

System 3

- The dog is **black**
- The dog is **white**

Contradictory

Singular

A system is non-singular only when every statement adds genuinely new information.

Adding a Third Statement

The same idea scales: with three statements about the dog, cat, and bird, we can still classify each system.

System 1

- Dog is **black**
- Cat is **orange**
- Bird is **red**

Complete

Non-singular

System 2

- Dog is **black**
- Dog is **black**
- Bird is **red**

Redundant

Singular

System 3

- Dog is **black**
- Dog is **black**
- Dog is **black**

Redundant

Singular

System 4

- Dog is **black**
- Dog is **white**
- Bird is **red**

Contradictory

Singular

From Statements to Rows to Rank

A Statement

“The dog is black”
carries one piece
of information.



A Row

Each statement
becomes one row
in a system / matrix.



Its Contribution

New info → independent row → rank
increases.
Repeated or conflicting info →
dependent row → rank stays the
same.

Rank = the number of statements (rows) that are truly independent — a system missing independent statements, or containing conflicting ones, is called singular.

DEFINITION

Rank of a Matrix

The rank of a matrix A is the maximum number of linearly independent rows (or columns) in A .

Denoted as: $\text{rank}(A)$

It equals the number of non-zero rows when A is written in row echelon form.

WHY RANK MATTERS

- Tells if a system of equations has a unique, infinite, or no solution
- Shows whether a matrix is invertible (full rank / non-singular)
- Used in image compression, computer graphics & machine learning
- Helps determine the dimension of the solution space

CONCEPT

Row Echelon Form (REF)

A matrix is in row echelon form when it satisfies all of the following:

- 1 All non-zero rows are above any rows of all zeros.
- 2 The leading (first non-zero) entry of each row — the pivot — is to the right of the pivot in the row above.
- 3 All entries in a column below a pivot are zero.

Example shape (★ = pivot, non-zero)

★	*	*
0	★	*
0	0	★

Elementary Row Operations

We use these operations to transform a matrix into echelon form without changing its rank.

1

Swap

Interchange two rows.

$$\mathbf{R}_i \leftrightarrow \mathbf{R}_j$$

2

Scale

Multiply a row by a non-zero constant.

$$\mathbf{R}_i \rightarrow k \cdot \mathbf{R}_i$$

3

Add / Subtract

Add a multiple of one row to another.

$$\mathbf{R}_i \rightarrow \mathbf{R}_i + k \cdot \mathbf{R}_j$$

THE KEY RULE

Rank = Number of Non-Zero Rows in Row Echelon Form

For a 3×3 matrix, the rank can be **0, 1, 2, or 3** depending on how many non-zero rows remain after reduction.

Rank 1

$$\begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank 2

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank 3

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{bmatrix}$$

WORKED EXAMPLE

Problem

Find the rank of the following 3×3 matrix using row echelon form:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

APPROACH

1. Use R1 as the pivot row (leading entry = 1).
2. Eliminate entries below the pivot in column 1.
3. Eliminate entries below the pivot in column 2.
4. Count the non-zero rows in the resulting echelon form.

STEP 1

Eliminate Below the First Pivot

Start:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

Result:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & -2 & 5 \end{bmatrix}$$

Row Operations Applied

$$\mathbf{R_2} \rightarrow \mathbf{R_2} - 2\mathbf{R_1}$$

$$(2, 5, 3) - 2(1, 2, 3) = (0, 1, -3)$$

$$\mathbf{R_3} \rightarrow \mathbf{R_3} - \mathbf{R_1}$$

$$(1, 0, 8) - (1, 2, 3) = (0, -2, 5)$$

STEP 2

Eliminate Below the Second Pivot

Start:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & -2 & 5 \end{bmatrix}$$

Row Echelon Form:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{bmatrix}$$

Row Operation Applied

$$\mathbf{R}_3 \rightarrow \mathbf{R}_3 + 2\mathbf{R}_2$$

$$(0, -2, 5) + 2(0, 1, -3) = (0, 0, -1)$$

This matrix is now in row echelon form — every pivot lies strictly right of the one above, and all entries below each pivot are zero.

RESULT

The Final Echelon Form

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{bmatrix}$$

All three rows are non-zero $\rightarrow$ 3 pivots.

Since a 3×3 matrix can have rank at most 3, this matrix has full rank.

$$\text{rank}(A) = 3$$

A is a full-rank (non-singular) matrix — its rows are linearly independent and A is invertible.

PRACTICE

Your Turn

Find the rank of matrix B using row echelon form. Reduce it step by step, then count the non-zero rows.

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

HINTS

- Notice Row 2 is exactly twice Row 1 — what will $R_2 - 2R_1$ give?
- One row will fully vanish (become all zeros).
- How many non-zero rows remain? That count is the rank.

Compressing Images by Reducing Rank

A grayscale image is just a matrix of pixel brightness values. Its rank tells us how much independent visual information it contains.

LOW-RANK APPROXIMATION

Instead of storing every pixel, we approximate the image matrix using only its top k most important row/column directions — a rank- k approximation.

As k grows, the approximation captures more detail — until it converges to the full-rank original.

WHY IT SAVES SPACE

A 200×200 image (rank 200) needs up to 40,000 numbers to store in full.

**A rank-10 approximation needs only
 $10 \times (200 + 200) = 4,000$ numbers**

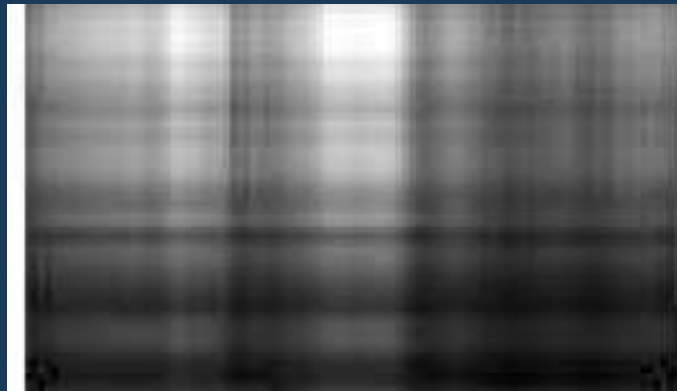
That's a 10× reduction in storage — the core idea behind many image and video compression techniques.

SEE IT IN ACTION

The Same Image at Different Ranks



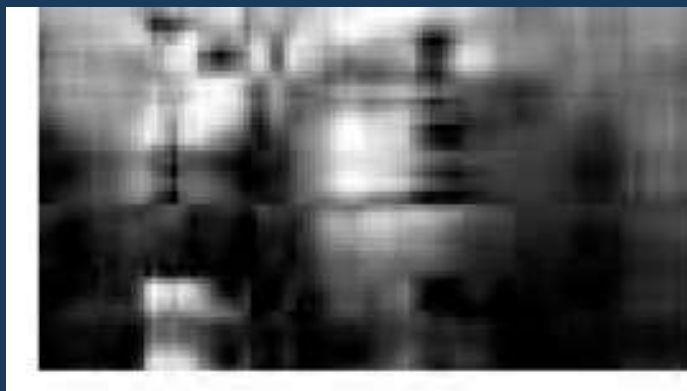
Original (Rank 200)



Rank 1



Rank 2



Rank 5



Rank 15



Rank 50

QUICK CHECK

Quiz: Systems of Statements

Given this system:

- Between the dog, the cat, and the bird, one is red.
- Between the dog and the cat, one is orange.
- The dog is black.

PROBLEM 1

What color is the bird?

PROBLEM 2

Is this system singular or non-singular?

SOLUTION

Working Through the Answer

Given

- Dog, cat, bird — one is **red**
- Dog, cat — one is **orange**
- The dog is **black**

SOLUTION 1

Since the dog is **black**, the “orange” statement must refer to the **cat**. That leaves “red” for the only remaining animal:

The bird is red.

SOLUTION 2

Each of the three statements gives new, independent information — none repeats or contradicts another.

The system is non-singular.

Takeaway: three animals, three genuinely independent facts $\rightarrow$ a 3×3 system with rank 3 — exactly like our matrix A from the worked example.

SUMMARY

Key Takeaways



A system is non-singular only when every statement / row adds independent information.



Rank = number of non-zero rows once a matrix is reduced to row echelon form.



Elementary row operations (swap, scale, add) reduce a matrix without changing its rank.



For our example, A reduced to 3 non-zero rows $\rightarrow \text{rank}(A) = 3$ (full rank, non-singular).



Reducing rank underlies real applications — like compressing images with far less stored data.

Practice with more 3x3 matrices to build speed and confidence!